

Лабораторная работа №15

Дисциплина: Администрирование сетевых подсистем

Бахи сиди али темассини

2026-02-16

1. Цель работы
2. Выполнение лабораторной работы
3. Выводы

Раздел 1

1. Цель работы

1.1 Цель работы

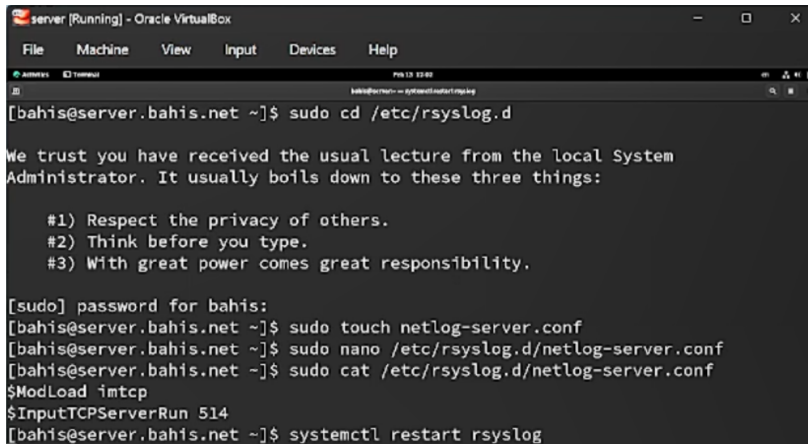
- Получение навыков по работе с журналами системных событий

Раздел 2

2. Выполнение лабораторной работы

2.1 Настройка сервера сетевого журнала

- Переход в каталог `/etc/rsyslog.d`



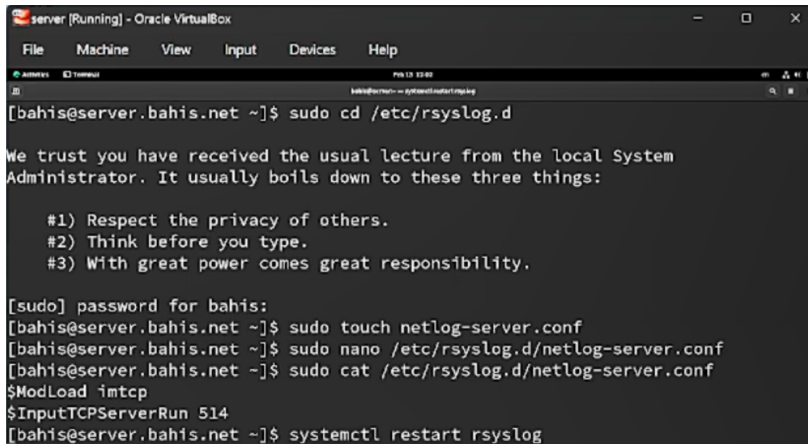
```
server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
bahis@server.bahis.net ~$ sudo cd /etc/rsyslog.d
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for bahis:
[bahis@server.bahis.net ~]$ sudo touch netlog-server.conf
[bahis@server.bahis.net ~]$ sudo nano /etc/rsyslog.d/netlog-server.conf
[bahis@server.bahis.net ~]$ sudo cat /etc/rsyslog.d/netlog-server.conf
$ModLoad imtcp
$InputTCPServerRun 514
[bahis@server.bahis.net ~]$ systemctl restart rsyslog
```

2.1 Настройка сервера сетевого журнала

- Переход в каталог `/etc/rsyslog.d`
- Создание файла `netlog-server.conf`



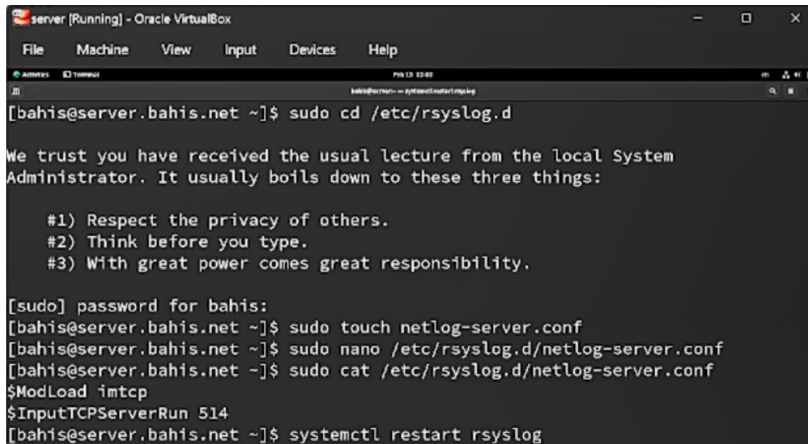
```
server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
bahis@server.bahis.net ~$ sudo cd /etc/rsyslog.d
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for bahis:
[bahis@server.bahis.net ~]$ sudo touch netlog-server.conf
[bahis@server.bahis.net ~]$ sudo nano /etc/rsyslog.d/netlog-server.conf
[bahis@server.bahis.net ~]$ sudo cat /etc/rsyslog.d/netlog-server.conf
$ModLoad imtcp
$InputTCPServerRun 514
[bahis@server.bahis.net ~]$ systemctl restart rsyslog
```

2.1 Настройка сервера сетевого журнала

- Переход в каталог `/etc/rsyslog.d`
- Создание файла `netlog-server.conf`
- Подготовка конфигурации для сетевого хранения журналов



```
server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
bahis@server.bahis.net ~$ sudo cd /etc/rsyslog.d
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for bahis:
[bahis@server.bahis.net ~]$ sudo touch netlog-server.conf
[bahis@server.bahis.net ~]$ sudo nano /etc/rsyslog.d/netlog-server.conf
[bahis@server.bahis.net ~]$ sudo cat /etc/rsyslog.d/netlog-server.conf
$ModLoad imtcp
$InputTCPServerRun 514
[bahis@server.bahis.net ~]$ systemctl restart rsyslog
```

- Включён приём журналов по TCP-порту 514

```
[bahis@server.bahis.net ~]$ sudo nano /etc/rsyslog.d/netlog-server.conf  
[bahis@server.bahis.net ~]$ sudo cat /etc/rsyslog.d/netlog-server.conf  
$ModLoad imtcp  
$InputTCPServerRun 514
```

Рисунок 2: Настройка приёма журналов по TCP 514 в rsyslog

- Включён приём журналов по TCP-порту 514
- Загружен модуль `imtcp`

```
[bahis@server.bahis.net ~]$ sudo nano /etc/rsyslog.d/netlog-server.conf  
[bahis@server.bahis.net ~]$ sudo cat /etc/rsyslog.d/netlog-server.conf  
$ModLoad imtcp  
$InputTCPServerRun 514
```

Рисунок 2: Настройка приёма журналов по TCP 514 в rsyslog

- Включён приём журналов по TCP-порту 514
- Загружен модуль `imtcp`
- Активирован сервер ввода `$InputTCPServerRun 514`

```
[bahis@server.bahis.net ~]$ sudo nano /etc/rsyslog.d/netlog-server.conf  
[bahis@server.bahis.net ~]$ sudo cat /etc/rsyslog.d/netlog-server.conf  
$ModLoad imtcp  
$InputTCPServerRun 514
```

Рисунок 2: Настройка приёма журналов по TCP 514 в rsyslog

- Перезапущена служба rsyslog

```
[bahis@server.bahis.net ~]$ sudo lsof | grep TCP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
cupsd      1697             root      7u      IPv6      2390
5          0t0      TCP localhost:ipp (LISTEN)
cupsd      1697             root      8u      IPv4      2390
6          0t0      TCP localhost:ipp (LISTEN)
sshd       1704             root      3u      IPv4      2389
4          0t0      TCP *:ssh (LISTEN)
sshd       1704             root      4u      IPv6      2390
9          0t0      TCP *:ssh (LISTEN)
sshd       24669            root      4u      IPv4      6053
2          0t0      TCP server.bahis.net:ssh->_gateway:58191 (ESTABLISHED)
sshd       24673            vagrant   4u      IPv4      6053
2          0t0      TCP server.bahis.net:ssh->_gateway:58191 (ESTABLISHED)
named      28059            named     22u      IPv4      7305
0          0t0      TCP localhost:domain (LISTEN)
named      28059            named     25u      IPv6      7305
2          0t0      TCP localhost:domain (LISTEN)
named      28059            named     26u      IPv4      7306
9          0t0      TCP localhost:rndc (LISTEN)
named      28059            named     27u      IPv6      7307
0          0t0      TCP localhost:rndc (LISTEN)
```

- Перезапущена служба `rsyslog`
- Выполнена проверка портов командой `lsof | grep TCP`

```
[bahis@server.bahis.net ~]$ sudo lsof | grep TCP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
cupsd      1697                root      7u      IPv6      2390
5          0t0      TCP localhost:ipp (LISTEN)
cupsd      1697                root      8u      IPv4      2390
6          0t0      TCP localhost:ipp (LISTEN)
sshd       1704                root      3u      IPv4      2389
4          0t0      TCP *:ssh (LISTEN)
sshd       1704                root      4u      IPv6      2390
9          0t0      TCP *:ssh (LISTEN)
sshd       24669               root      4u      IPv4      6053
2          0t0      TCP server.bahis.net:ssh->_gateway:58191 (ESTABLISHED)
sshd       24673               vagrant   4u      IPv4      6053
2          0t0      TCP server.bahis.net:ssh->_gateway:58191 (ESTABLISHED)
named      28059               named     22u      IPv4      7305
0          0t0      TCP localhost:domain (LISTEN)
named      28059               named     25u      IPv6      7305
2          0t0      TCP localhost:domain (LISTEN)
named      28059               named     26u      IPv4      7306
9          0t0      TCP localhost:rndc (LISTEN)
named      28059               named     27u      IPv6      7307
0          0t0      TCP localhost:rndc (LISTEN)
```

- Перезапущена служба rsyslog
- Выполнена проверка портов командой `lsof | grep TCP`
- Подтверждено состояние LISTEN для rsyslog

```
[bahis@server.bahis.net ~]$ sudo lsof | grep TCP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
cupsd      1697          root      7u      IPv6      2390
5          0t0      TCP localhost:ipp (LISTEN)
cupsd      1697          root      8u      IPv4      2390
6          0t0      TCP localhost:ipp (LISTEN)
sshd       1704          root      3u      IPv4      2389
4          0t0      TCP *:ssh (LISTEN)
sshd       1704          root      4u      IPv6      2390
9          0t0      TCP *:ssh (LISTEN)
sshd       24669         root      4u      IPv4      6053
2          0t0      TCP server.bahis.net:ssh->_gateway:58191 (ESTABLISHED)
sshd       24673         vagrant   4u      IPv4      6053
2          0t0      TCP server.bahis.net:ssh->_gateway:58191 (ESTABLISHED)
named      28059         named     22u      IPv4      7305
0          0t0      TCP localhost:domain (LISTEN)
named      28059         named     25u      IPv6      7305
2          0t0      TCP localhost:domain (LISTEN)
named      28059         named     26u      IPv4      7306
9          0t0      TCP localhost:rndc (LISTEN)
named      28059         named     27u      IPv6      7307
0          0t0      TCP localhost:rndc (LISTEN)
```

- Открыт TCP-порт 514 в `firewalld`

```
[bahis@server.bahis.net ~]$ sudo firewall-cmd --add-port=514/tcp  
success  
[bahis@server.bahis.net ~]$ sudo firewall-cmd --add-port=514/tcp --permanent  
success  
[bahis@server.bahis.net ~]$
```

Рисунок 4: Настройка `firewalld` для открытия TCP-порта 514

- Открыт TCP-порт 514 в `firewalld`
- Добавлено временное и постоянное правило

```
[bahis@server.bahis.net ~]$ sudo firewall-cmd --add-port=514/tcp  
success  
[bahis@server.bahis.net ~]$ sudo firewall-cmd --add-port=514/tcp --permanent  
success  
[bahis@server.bahis.net ~]$
```

Рисунок 4: Настройка `firewalld` для открытия TCP-порта 514

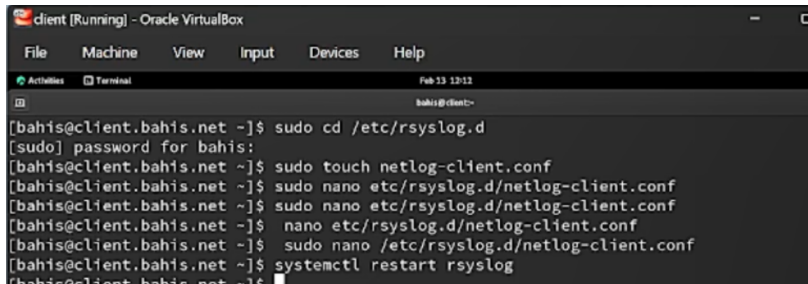
- Открыт TCP-порт 514 в `firewalld`
- Добавлено временное и постоянное правило
- Разрешён приём сетевых сообщений журнала

```
[bahis@server.bahis.net ~]$ sudo firewall-cmd --add-port=514/tcp  
success  
[bahis@server.bahis.net ~]$ sudo firewall-cmd --add-port=514/tcp --permanent  
success  
[bahis@server.bahis.net ~]$
```

Рисунок 4: Настройка `firewalld` для открытия TCP-порта 514

2.2 Настройка клиента сетевого журнала

- Переход в каталог `/etc/rsyslog.d`

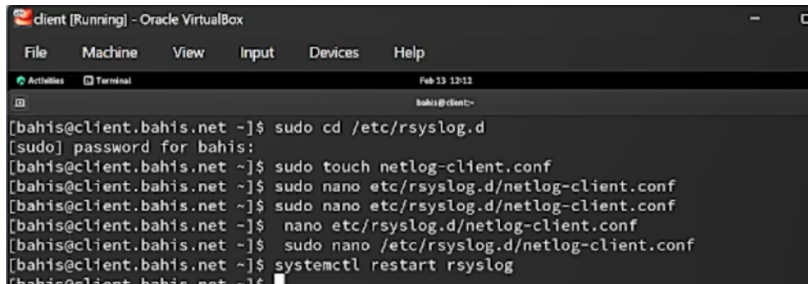


```
client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 12:12
bahis@client~
[bahis@client.bahis.net ~]$ sudo cd /etc/rsyslog.d
[sudo] password for bahis:
[bahis@client.bahis.net ~]$ sudo touch netlog-client.conf
[bahis@client.bahis.net ~]$ sudo nano etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ sudo nano etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ nano etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ sudo nano /etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ systemctl restart rsyslog
[bahis@client.bahis.net ~]$
```

Рисунок 5: Создание файла конфигурации `netlog-client.conf` на клиенте

2.2 Настройка клиента сетевого журнала

- Переход в каталог `/etc/rsyslog.d`
- Создание файла `netlog-client.conf`

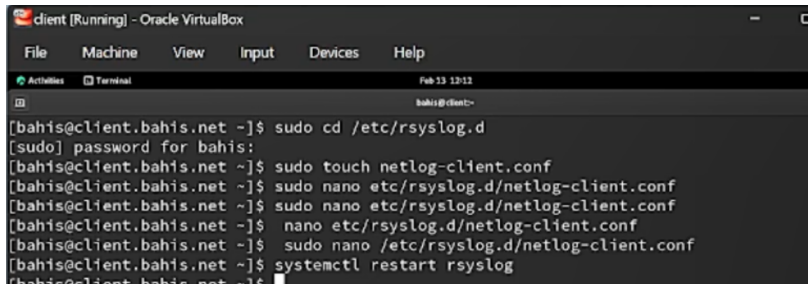


```
client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 12:12
bahis@client~
[bahis@client.bahis.net ~]$ sudo cd /etc/rsyslog.d
[sudo] password for bahis:
[bahis@client.bahis.net ~]$ sudo touch netlog-client.conf
[bahis@client.bahis.net ~]$ sudo nano etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ sudo nano etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ nano etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ sudo nano /etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ systemctl restart rsyslog
[bahis@client.bahis.net ~]$
```

Рисунок 5: Создание файла конфигурации `netlog-client.conf` на клиенте

2.2 Настройка клиента сетевого журнала

- Переход в каталог `/etc/rsyslog.d`
- Создание файла `netlog-client.conf`
- Подготовка конфигурации отправки журналов



```
client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 12:12
bahis@client~
[bahis@client.bahis.net ~]$ sudo cd /etc/rsyslog.d
[sudo] password for bahis:
[bahis@client.bahis.net ~]$ sudo touch netlog-client.conf
[bahis@client.bahis.net ~]$ sudo nano etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ sudo nano etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ nano etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ sudo nano /etc/rsyslog.d/netlog-client.conf
[bahis@client.bahis.net ~]$ systemctl restart rsyslog
[bahis@client.bahis.net ~]$
```

Рисунок 5: Создание файла конфигурации `netlog-client.conf` на клиенте

- Настроено перенаправление *.* @@server.bahis.net:514

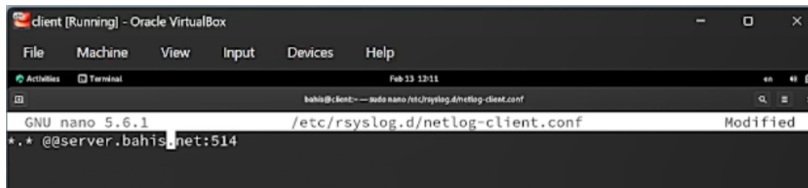


Рисунок 6: Настройка перенаправления журналов на сервер по TCP 514

- Настроено перенаправление * . * @@server.bahis.net:514
- Использован протокол TCP (@@)

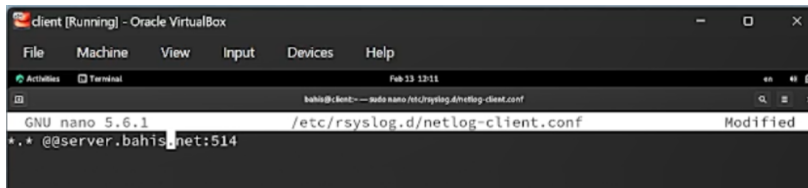
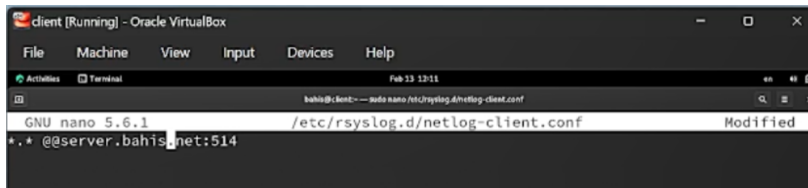


Рисунок 6: Настройка перенаправления журналов на сервер по TCP 514

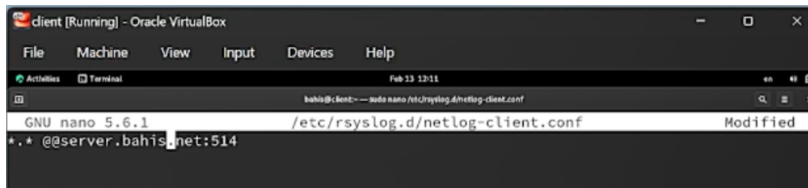
- Настроено перенаправление * . * @@server.bahis.net:514
- Использован протокол TCP (@@)
- Организована передача всех сообщений на сервер



The screenshot shows a terminal window titled "client [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". Below the menu bar, there is a status bar showing "Feb 13 12:11". The terminal content shows the command `sudo nano /etc/rsyslog.d/netlog-client.conf` being executed. The nano editor interface displays the file `/etc/rsyslog.d/netlog-client.conf` as "Modified". The content of the file is `*.* @@server.bahis.net:514`.

Рисунок 6: Настройка перенаправления журналов на сервер по TCP 514

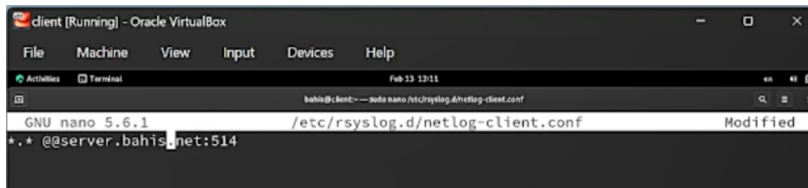
- Настроено перенаправление * . * @@server.bahis.net:514
- Использован протокол TCP (@@)
- Организована передача всех сообщений на сервер
- Перезапущена служба rsyslog



```
client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Feb 13 12:11
bahis@client:~$ sudo nano /etc/rsyslog.d/netlog-client.conf
GNU nano 5.6.1 /etc/rsyslog.d/netlog-client.conf Modified
*.* @@server.bahis.net:514
```

Рисунок 6: Настройка перенаправления журналов на сервер по TCP 514

- Настроено перенаправление * . * @@server.bahis.net:514
- Использован протокол TCP (@@)
- Организована передача всех сообщений на сервер
- Перезапущена служба rsyslog
- Применена новая конфигурация клиента

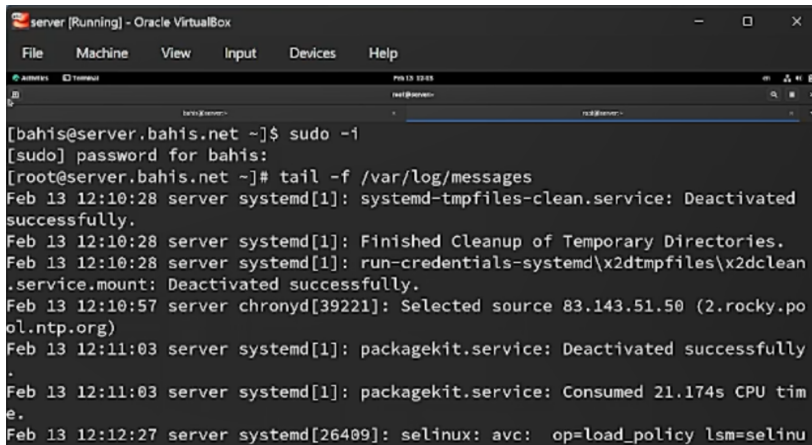


```
client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Feb 13 12:11
bahis@client:~$ sudo nano /etc/rsyslog.d/netlog-client.conf
GNU nano 5.6.1 /etc/rsyslog.d/netlog-client.conf Modified
*.* @@server.bahis.net:514
```

Рисунок 6: Настройка перенаправления журналов на сервер по TCP 514

2.3 Просмотр журнала

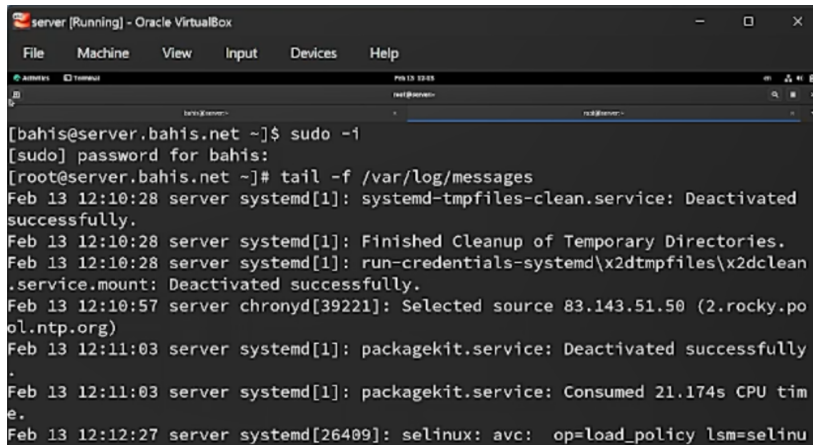
- Выполнен просмотр `/var/log/messages` с помощью `tail -f`



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 12:05
[bahtis@server.bahis.net ~]$ sudo -i
[sudo] password for bahtis:
[root@server.bahis.net ~]# tail -f /var/log/messages
Feb 13 12:10:28 server systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Feb 13 12:10:28 server systemd[1]: Finished Cleanup of Temporary Directories.
Feb 13 12:10:28 server systemd[1]: run-credentials-systemd\x2dtmpfiles\x2dclean.service.mount: Deactivated successfully.
Feb 13 12:10:57 server chronyd[39221]: Selected source 83.143.51.50 (2.rocky.pool.ntp.org)
Feb 13 12:11:03 server systemd[1]: packagekit.service: Deactivated successfully.
Feb 13 12:11:03 server systemd[1]: packagekit.service: Consumed 21.174s CPU time.
Feb 13 12:12:27 server systemd[26409]: selinux: avc: op=load_policy lsm=selinu
```

2.3 Просмотр журнала

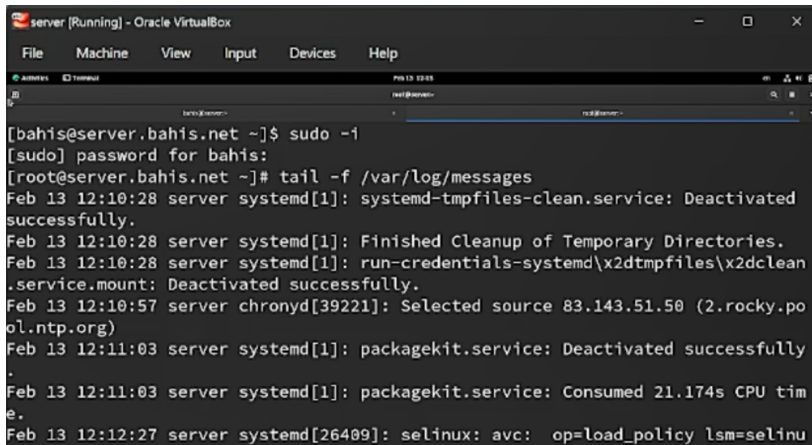
- Выполнен просмотр `/var/log/messages` с помощью `tail -f`
- Отображаются записи `server` и `client`



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 12:05
[bahts@server.bahis.net ~]$ sudo -i
[sudo] password for bahts:
[root@server.bahis.net ~]# tail -f /var/log/messages
Feb 13 12:10:28 server systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Feb 13 12:10:28 server systemd[1]: Finished Cleanup of Temporary Directories.
Feb 13 12:10:28 server systemd[1]: run-credentials-systemd\x2dtmpfiles\x2dclean.service.mount: Deactivated successfully.
Feb 13 12:10:57 server chronyd[39221]: Selected source 83.143.51.50 (2.rocky.pool.ntp.org)
Feb 13 12:11:03 server systemd[1]: packagekit.service: Deactivated successfully.
Feb 13 12:11:03 server systemd[1]: packagekit.service: Consumed 21.174s CPU time.
Feb 13 12:12:27 server systemd[26409]: selinux: avc: op=load_policy lsm=selinu
```

2.3 Просмотр журнала

- Выполнен просмотр `/var/log/messages` с помощью `tail -f`
- Отображаются записи `server` и `client`
- Подтверждён приём удалённых журналов



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 12:05
[bahtis@server.bahis.net ~]$ sudo -i
[sudo] password for bahtis:
[root@server.bahis.net ~]# tail -f /var/log/messages
Feb 13 12:10:28 server systemd[1]: systemd-tmpfiles-clean.service: Deactivated
successfully.
Feb 13 12:10:28 server systemd[1]: Finished Cleanup of Temporary Directories.
Feb 13 12:10:28 server systemd[1]: run-credentials-systemd\x2dtmpfiles\x2dclean
.service.mount: Deactivated successfully.
Feb 13 12:10:57 server chronyd[39221]: Selected source 83.143.51.50 (2.rocky.po
ol.ntp.org)
Feb 13 12:11:03 server systemd[1]: packagekit.service: Deactivated successfully
.
Feb 13 12:11:03 server systemd[1]: packagekit.service: Consumed 21.174s CPU tim
e.
Feb 13 12:12:27 server systemd[26409]: selinux: avc: op=load_policy lsm=selinu
```

- Запущена программа `gnome-system-monitor`

[illegible]

- Запущена программа `gnome-system-monitor`
- Отображены процессы, PID, использование CPU и памяти

[illegible]

- Установлен пакет lnav через dnf

```
[bahis@server.bahis.net ~]$ sudo dnf -y install lnav
[sudo] password for bahis:
Last metadata expiration check: 0:17:49 ago on Fri 13 Feb 2026 12:01:29 PM UTC.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing:
lnav                    x86_64            0.11.1-1.el9     epel               2.4 M
=====
Transaction Summary
=====
Install 1 Package

Total download size: 2.4 M
Installed size: 6.1 M
```

Рисунок 9: Установка lnav на сервере

- Установлен пакет lnav через dnf
- Подтверждена успешная установка версии 0.11.1-1.el9

```
[bahis@server.bahis.net ~]$ sudo dnf -y install lnav
[sudo] password for bahis:
Last metadata expiration check: 0:17:49 ago on Fri 13 Feb 2026 12:01:29 PM UTC.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing:
lnav                    x86_64            0.11.1-1.el9     epel              2.4 M
=====
Transaction Summary
=====
Install 1 Package

Total download size: 2.4 M
Installed size: 6.1 M
```

Рисунок 9: Установка lnav на сервере

- Выполнен просмотр журналов с помощью `lnav`

```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
server [Running] - Oracle VirtualBox
LOG
2026-02-13T12:26:57 UTC Press ENTER to focus on the breadcrumb bar
LOG 2026-02-13T12:05:23.000 syslog_log messages[5,094] kernel
Feb 13 12:05:23 server kernel: RPC: Registered tcp NFSv4.1 backchannel transport module.
Feb 13 12:05:23 server systemd[1]: Mounted RPC Pipe File System.
Feb 13 12:05:23 server systemd[1]: Reached target rpc_pipefs.target.
Feb 13 12:05:23 server systemd[1]: RPC security service for NFS client and server was skipped
Feb 13 12:05:23 server systemd[1]: Reached target NFS client services.
Feb 13 12:05:23 server systemd[1]: Reached target Preparation for Remote File Systems.
Feb 13 12:05:23 server systemd[1]: Stopping GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: gssproxy.service: Deactivated successfully.
Feb 13 12:05:23 server systemd[1]: Stopped GSSAPI Proxy Daemon.
Feb 13 12:05:23 server rsyslogd[28612]: imjournal: journal files changed, reloading... [v8.2
Feb 13 12:05:23 server rsyslogd[28612]: imjournal: journal files changed, reloading... [v8.2
Feb 13 12:05:23 server systemd[1]: Starting GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: Started GSSAPI Proxy Daemon.
Feb 13 12:05:23 server systemd[1]: Stopping GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: gssproxy.service: Deactivated successfully.
Feb 13 12:05:23 server systemd[1]: Stopped GSSAPI Proxy Daemon.
Feb 13 12:05:23 server systemd[1]: Starting GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: Started GSSAPI Proxy Daemon.
Feb 13 12:05:24 server systemd[1]: Started /usr/bin/systemctl start man-db-cache-update.
Feb 13 12:05:24 server systemd[1]: Starting man-db-cache-update.service...
Feb 13 12:05:24 server systemd[1]: Reloading.
Feb 13 12:05:24 server systemd-rc-local-generator: /etc/rc.d/rc.local is not marked executabl
Feb 13 12:05:24 server systemd[1]: Queuing reload/restart jobs for marked units..
Feb 13 12:05:29 server systemd[1]: man-db-cache-update.service: Deactivated successfully.
Feb 13 12:05:29 server systemd[1]: Finished man-db-cache-update.service.
Feb 13 12:05:29 server systemd[1]: man-db-cache-update.service: Consumed 2.340s CPU time.
Feb 13 12:05:29 server systemd[1]: run-ra751ed17e857419585e859b8666c9bbb.service: Deactivated
Feb 13 12:05:30 server kernel: SELinux: Converting 729 SID table entries...
Feb 13 12:05:30 server kernel: SELinux: policy capability network_peer_controls=1
Feb 13 12:05:30 server kernel: SELinux: policy capability open_perms=1
```

- Выполнен просмотр журналов с помощью `lnav`
- Отображаются записи сервера и клиента

```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
server [Running] - Oracle VirtualBox
Feb 13 12:05:23 server kernel: RPC: Registered tcp NFSv4.1 backchannel transport module.
Feb 13 12:05:23 server systemd[1]: Mounted RPC Pipe File System.
Feb 13 12:05:23 server systemd[1]: Reached target rpc_pipefs.target.
Feb 13 12:05:23 server systemd[1]: RPC security service for NFS client and server was skipped
Feb 13 12:05:23 server systemd[1]: Reached target NFS client services.
Feb 13 12:05:23 server systemd[1]: Reached target Preparation for Remote File Systems.
Feb 13 12:05:23 server systemd[1]: Stopping GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: gssproxy.service: Deactivated successfully.
Feb 13 12:05:23 server systemd[1]: Stopped GSSAPI Proxy Daemon.
Feb 13 12:05:23 server rsyslogd[28612]: imjournal: journal files changed, reloading... [v8.2
Feb 13 12:05:23 server rsyslogd[28612]: imjournal: journal files changed, reloading... [v8.2
Feb 13 12:05:23 server systemd[1]: Starting GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: Started GSSAPI Proxy Daemon.
Feb 13 12:05:23 server systemd[1]: Stopping GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: gssproxy.service: Deactivated successfully.
Feb 13 12:05:23 server systemd[1]: Stopped GSSAPI Proxy Daemon.
Feb 13 12:05:23 server systemd[1]: Starting GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: Started GSSAPI Proxy Daemon.
Feb 13 12:05:24 server systemd[1]: Started /usr/bin/systemctl start man-db-cache-update.
Feb 13 12:05:24 server systemd[1]: Starting man-db-cache-update.service...
Feb 13 12:05:24 server systemd[1]: Reloading.
Feb 13 12:05:24 server systemd-rc-local-generator: /etc/rc.d/rc.local is not marked executabl
Feb 13 12:05:24 server systemd[1]: Queuing reload/restart jobs for marked units..
Feb 13 12:05:29 server systemd[1]: man-db-cache-update.service: Deactivated successfully.
Feb 13 12:05:29 server systemd[1]: Finished man-db-cache-update.service.
Feb 13 12:05:29 server systemd[1]: man-db-cache-update.service: Consumed 2.340s CPU time.
Feb 13 12:05:29 server systemd[1]: run-ra751ed17e857419585e859b8666c9bbb.service: Deactivated
Feb 13 12:05:30 server kernel: SELinux: Converting 729 SID table entries...
Feb 13 12:05:30 server kernel: SELinux: policy capability network_peer_controls=1
Feb 13 12:05:30 server kernel: SELinux: policy capability open_perms=1
```

- Выполнен просмотр журналов с помощью `lnav`
- Отображаются записи сервера и клиента
- Видны сообщения `systemd`, `rsyslog`, `cupsd`

```

server [Running] - Oracle VirtualBox
File Machine View Input Devices Help

LOG > 2026-02-13T12:05:23.000 > syslog_log > messages[5,094] > kernel
2026-02-13T12:26:57 UTC Press ENTER to focus on the breadcrumb bar
Feb 13 12:05:23 server kernel: RPC: Registered tcp NFSv4.1 backchannel transport module.
Feb 13 12:05:23 server systemd[1]: Mounted RPC Pipe File System.
Feb 13 12:05:23 server systemd[1]: Reached target rpc_pipefs.target.
Feb 13 12:05:23 server systemd[1]: RPC security service for NFS client and server was skipped
Feb 13 12:05:23 server systemd[1]: Reached target NFS client services.
Feb 13 12:05:23 server systemd[1]: Reached target Preparation for Remote File Systems.
Feb 13 12:05:23 server systemd[1]: Stopping GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: gssproxy.service: Deactivated successfully.
Feb 13 12:05:23 server systemd[1]: Stopped GSSAPI Proxy Daemon.
Feb 13 12:05:23 server rsyslogd[28612]: imjournal: journal files changed, reloading... [v8.2
Feb 13 12:05:23 server rsyslogd[28612]: imjournal: journal files changed, reloading... [v8.2
Feb 13 12:05:23 server systemd[1]: Starting GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: Started GSSAPI Proxy Daemon.
Feb 13 12:05:23 server systemd[1]: Stopping GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: gssproxy.service: Deactivated successfully.
Feb 13 12:05:23 server systemd[1]: Stopped GSSAPI Proxy Daemon.
Feb 13 12:05:23 server systemd[1]: Starting GSSAPI Proxy Daemon...
Feb 13 12:05:23 server systemd[1]: Started GSSAPI Proxy Daemon.
Feb 13 12:05:24 server systemd[1]: Started /usr/bin/systemctl start man-db-cache-update.
Feb 13 12:05:24 server systemd[1]: Starting man-db-cache-update.service...
Feb 13 12:05:24 server systemd[1]: Reloading.
Feb 13 12:05:24 server systemd-rc-local-generator: /etc/rc.d/rc.local is not marked executabl
Feb 13 12:05:24 server systemd[1]: Queuing reload/restart jobs for marked units..
Feb 13 12:05:29 server systemd[1]: man-db-cache-update.service: Deactivated successfully.
Feb 13 12:05:29 server systemd[1]: Finished man-db-cache-update.service.
Feb 13 12:05:29 server systemd[1]: man-db-cache-update.service: Consumed 2.340s CPU time.
Feb 13 12:05:29 server systemd[1]: run-ra751ed17e857419585e859b8666c9bbb.service: Deactivated
Feb 13 12:05:30 server kernel: SELinux: Converting 729 SID table entries...
Feb 13 12:05:30 server kernel: SELinux: policy capability network_peer_controls=1
Feb 13 12:05:30 server kernel: SELinux: policy capability open_perms=1
  
```


- Выполнен просмотр журналов клиента через Inav

```
client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Feb 11 16:44 en
LOG
2026-02-11T16:44:05 UTC Press ENTER to focus on the breadcrumb bar
LOG 2026-02-11T16:04:47.000 syslog_log messages[25,534]
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:50 client systemd[4701]: Started Application launched by gnome-shell.
Feb 11 16:04:50 client systemd[4701]: Created slice /app/org.gnome.Terminal.
Feb 11 16:04:50 client systemd[4701]: Starting GNOME Terminal Server...
Feb 11 16:04:50 client systemd[4701]: Started GNOME Terminal Server.
Feb 11 16:04:50 client systemd[4701]: Started VTE child process 5618 launched by gnome-terminal-server
Feb 11 16:04:56 client gnome-shell[4873]: Can't update stage views actor MetaWindowGroup is on because
Feb 11 16:04:56 client gnome-shell[4873]: Can't update stage views actor MetaWindowActorX11 is on because
Feb 11 16:04:56 client gnome-shell[4873]: Can't update stage views actor MetaSurfaceActorX11 is on because
Feb 11 16:05:21 client systemd[4701]: Starting Mark boot as successful...
Feb 11 16:05:21 client systemd[4701]: Finished Mark boot as successful.
Feb 11 16:08:21 client systemd[4701]: Created slice User Background Tasks Slice.
Feb 11 16:08:21 client systemd[4701]: Starting Cleanup of User's Temporary Files and Directories...
Feb 11 16:08:21 client systemd[4701]: Finished Cleanup of User's Temporary Files and Directories.
Feb 11 16:08:49 client systemd[1]: packagekit.service: Deactivated successfully.
Feb 11 16:08:49 client systemd[1]: packagekit.service: Consumed 4.054s CPU time.
Feb 11 16:18:21 client systemd[1]: Starting Cleanup of Temporary Directories...
Feb 11 16:18:21 client systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Feb 11 16:18:21 client systemd[1]: Finished Cleanup of Temporary Directories.
Feb 11 16:18:21 client systemd[1]: run-credentials-systemd\x2dtmpfiles\x2dclean.service.mount: Deactivated successfully.
Feb 11 16:22:21 client systemd[1]: Starting dnf makecache...
Feb 11 16:22:22 client dnf[5728]: Metadata cache refreshed recently.
Feb 11 16:22:22 client systemd[1]: dnf-makecache.service: Deactivated successfully.
Feb 11 16:22:22 client systemd[1]: Finished dnf makecache.
Feb 11 16:37:24 client systemd[1]: Stopping System Logging Service...
Feb 11 16:37:25 client rsyslogd[1020]: [origin software="rsyslogd" swVersion="8.2506.0-2.el9" x-pid="1020"]
Feb 11 16:37:25 client systemd[1]: rsyslog.service: Deactivated successfully.
Feb 11 16:37:25 client systemd[1]: Stopped System Logging Service.
Feb 11 16:37:25 client systemd[1]: Starting System Logging Service...
Feb 11 16:37:25 client systemd[1]: Started System Logging Service.
Feb 11 16:37:25 client rsyslogd[5815]: forain software="rsyslogd" swVersion="8.2506.0-2.el9" x-pid="5815"
```

- Выполнен просмотр журналов клиента через Inav
- Отображаются записи systemd, rsyslog, packagekit

```
client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Feb 11 16:44
LOG
2026-02-11T16:44:05 UTC Press ENTER to focus on the breadcrumb bar
LOG | 2026-02-11T16:04:47.000|syslog_log|messages[25,534]|
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtuall: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtuall: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtuall: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtuall: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtuall: unable to get EDID for
Feb 11 16:04:50 client systemd[4701]: Started Application launched by gnome-shell.
Feb 11 16:04:50 client systemd[4701]: Created slice /app/org.gnome.Terminal.
Feb 11 16:04:50 client systemd[4701]: Starting GNOME Terminal Server...
Feb 11 16:04:50 client systemd[4701]: Started GNOME Terminal Server.
Feb 11 16:04:50 client systemd[4701]: Started VTE child process 5618 launched by gnome-terminal-server
Feb 11 16:04:56 client gnome-shell[4873]: Can't update stage views actor MetaWindowGroup is on because
Feb 11 16:04:56 client gnome-shell[4873]: Can't update stage views actor MetaWindowActorX11 is on because
Feb 11 16:04:56 client gnome-shell[4873]: Can't update stage views actor MetaSurfaceActorX11 is on because
Feb 11 16:05:21 client systemd[4701]: Starting Mark boot as successful...
Feb 11 16:05:21 client systemd[4701]: Finished Mark boot as successful.
Feb 11 16:08:21 client systemd[4701]: Created slice User Background Tasks Slice.
Feb 11 16:08:21 client systemd[4701]: Starting Cleanup of User's Temporary Files and Directories...
Feb 11 16:08:21 client systemd[4701]: Finished Cleanup of User's Temporary Files and Directories.
Feb 11 16:08:49 client systemd[1]: packagekit.service: Deactivated successfully.
Feb 11 16:08:49 client systemd[1]: packagekit.service: Consumed 4.054s CPU time.
Feb 11 16:18:21 client systemd[1]: Starting Cleanup of Temporary Directories...
Feb 11 16:18:21 client systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Feb 11 16:18:21 client systemd[1]: Finished Cleanup of Temporary Directories.
Feb 11 16:18:21 client systemd[1]: run-credentials-systemd\x2dtmpfiles\x2dclean.service.mount: Deactivated successfully.
Feb 11 16:22:21 client systemd[1]: Starting dnf makecache...
Feb 11 16:22:22 client dnf[5728]: Metadata cache refreshed recently.
Feb 11 16:22:22 client systemd[1]: dnf-makecache.service: Deactivated successfully.
Feb 11 16:22:22 client systemd[1]: Finished dnf makecache.
Feb 11 16:37:24 client systemd[1]: Stopping System Logging Service...
Feb 11 16:37:25 client rsyslogd[1020]: [origin software="rsyslogd" swVersion="8.2506.0-2.el9" x-pid="1020"]
Feb 11 16:37:25 client systemd[1]: rsyslog.service: Deactivated successfully.
Feb 11 16:37:25 client systemd[1]: Stopped System Logging Service.
Feb 11 16:37:25 client systemd[1]: Starting System Logging Service...
Feb 11 16:37:25 client systemd[1]: Started System Logging Service.
Feb 11 16:37:25 client rsyslogd[5815]: forain software="rsyslogd" swVersion="8.2506.0-2.el9" x-pid="5815"
```

- Выполнен просмотр журналов клиента через Inav
- Отображаются записи systemd, rsyslog, packagekit
- Подтверждена корректная работа логирования

The screenshot shows a terminal window titled "client [Running] - Oracle VirtualBox". The terminal displays a log of system events. The log starts with a timestamp "2026-02-11T16:44:05 UTC" and a message "LOG [2026-02-11T16:04:47.000] syslog_log[messages[25,534]]". The log entries include messages from "gsd-color[5151]" about EDID, "systemd[4701]" about starting GNOME Terminal Server and VTE child process, "gnome-shell[4873]" about stage views, "systemd[4701]" about Mark boot, "systemd[4701]" about Cleanup of User's Temporary Files and Directories, "packagekit.service", "dnf[5728]" about metadata cache, and "rsyslogd[1020]" about stopping and starting System Logging Service.

```

client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Feb 11 16:44 en
LOG
2026-02-11T16:44:05 UTC Press ENTER to focus on the breadcrumb bar
LOG [2026-02-11T16:04:47.000] syslog_log[messages[25,534]]
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:47 client gsd-color[5151]: unable to get EDID for xrandr-Virtual1: unable to get EDID for
Feb 11 16:04:50 client systemd[4701]: Started Application launched by gnome-shell.
Feb 11 16:04:50 client systemd[4701]: Created slice /app/org.gnome.Terminal.
Feb 11 16:04:50 client systemd[4701]: Starting GNOME Terminal Server...
Feb 11 16:04:50 client systemd[4701]: Started GNOME Terminal Server.
Feb 11 16:04:50 client systemd[4701]: Started VTE child process 5618 launched by gnome-terminal-server
Feb 11 16:04:56 client gnome-shell[4873]: Can't update stage views actor MetaWindowGroup is on because
Feb 11 16:04:56 client gnome-shell[4873]: Can't update stage views actor MetaWindowActorX11 is on because
Feb 11 16:04:56 client gnome-shell[4873]: Can't update stage views actor MetaSurfaceActorX11 is on because
Feb 11 16:05:21 client systemd[4701]: Starting Mark boot as successful...
Feb 11 16:05:21 client systemd[4701]: Finished Mark boot as successful.
Feb 11 16:08:21 client systemd[4701]: Created slice User Background Tasks Slice.
Feb 11 16:08:21 client systemd[4701]: Starting Cleanup of User's Temporary Files and Directories...
Feb 11 16:08:21 client systemd[4701]: Finished Cleanup of User's Temporary Files and Directories.
Feb 11 16:08:49 client systemd[1]: packagekit.service: Deactivated successfully.
Feb 11 16:08:49 client systemd[1]: packagekit.service: Consumed 4.054s CPU time.
Feb 11 16:18:21 client systemd[1]: Starting Cleanup of Temporary Directories...
Feb 11 16:18:21 client systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Feb 11 16:18:21 client systemd[1]: Finished Cleanup of Temporary Directories.
Feb 11 16:18:21 client systemd[1]: run-credentials-systemd\x2dtmpfiles\x2dclean.service.mount: Deactivated successfully.
Feb 11 16:22:21 client systemd[1]: Starting dnf makecache...
Feb 11 16:22:22 client dnf[5728]: Metadata cache refreshed recently.
Feb 11 16:22:22 client systemd[1]: dnf-makecache.service: Deactivated successfully.
Feb 11 16:22:22 client systemd[1]: Finished dnf makecache.
Feb 11 16:37:24 client systemd[1]: Stopping System Logging Service...
Feb 11 16:37:25 client rsyslogd[1020]: [origin software="rsyslogd" swVersion="8.2506.0-2.el9" x-pid="1020"]
Feb 11 16:37:25 client systemd[1]: rsyslog.service: Deactivated successfully.
Feb 11 16:37:25 client systemd[1]: Stopped System Logging Service.
Feb 11 16:37:25 client systemd[1]: Starting System Logging Service...
Feb 11 16:37:25 client systemd[1]: Started System Logging Service.
Feb 11 16:37:25 client rsyslogd[5815]: forain software="rsyslogd" swVersion="8.2506.0-2.el9" x-pid="5815"

```

2.4 Внесение изменений в настройки внутреннего

- Переход в `/vagrant/provision/server`

```
[bahis@server.bahis.net ~]$ cd /vagrant/provision/server
bash: cd: /vagrant/provision/server: No such file or directory
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# cd /vagrant/provision/server
-bash: cd: /vagrant/provision/server: No such file or directory
[root@server.bahis.net ~]# mkdir -p /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.bahis.net ~]# cp -R /etc/rsyslog.d/netlog-server.conf /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch netlog.sh
[root@server.bahis.net server]# chmod +x netlog.sh
[root@server.bahis.net server]# nano netlog.sh
```

Рисунок 12: Создание каталога netlog и копирование конфигурации rsyslog

2.4 Внесение изменений в настройки внутреннего

- Переход в `/vagrant/provision/server`
- Создание каталога `netlog/etc/rsyslog.d`

```
[bahis@server.bahis.net ~]$ cd /vagrant/provision/server
bash: cd: /vagrant/provision/server: No such file or directory
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# cd /vagrant/provision/server
-bash: cd: /vagrant/provision/server: No such file or directory
[root@server.bahis.net ~]# mkdir -p /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.bahis.net ~]# cp -R /etc/rsyslog.d/netlog-server.conf /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch netlog.sh
[root@server.bahis.net server]# chmod +x netlog.sh
[root@server.bahis.net server]# nano netlog.sh
```

Рисунок 12: Создание каталога netlog и копирование конфигурации rsyslog

2.4 Внесение изменений в настройки внутреннего

- Переход в `/vagrant/provision/server`
- Создание каталога `netlog/etc/rsyslog.d`
- Копирование `netlog-server.conf`

```
[bahis@server.bahis.net ~]$ cd /vagrant/provision/server
bash: cd: /vagrant/provision/server: No such file or directory
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# cd /vagrant/provision/server
-bash: cd: /vagrant/provision/server: No such file or directory
[root@server.bahis.net ~]# mkdir -p /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.bahis.net ~]# cp -R /etc/rsyslog.d/netlog-server.conf /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch netlog.sh
[root@server.bahis.net server]# chmod +x netlog.sh
[root@server.bahis.net server]# nano netlog.sh
```

Рисунок 12: Создание каталога netlog и копирование конфигурации rsyslog

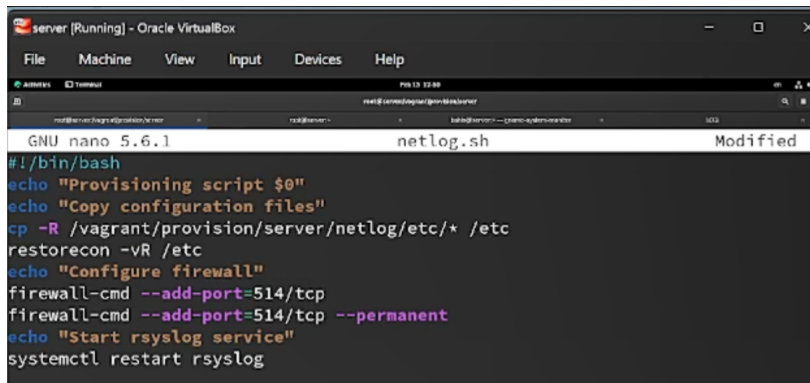
2.4 Внесение изменений в настройки внутреннего

- Переход в `/vagrant/provision/server`
- Создание каталога `netlog/etc/rsyslog.d`
- Копирование `netlog-server.conf`
- Подготовка структуры provisioning

```
[bahis@server.bahis.net ~]$ cd /vagrant/provision/server
bash: cd: /vagrant/provision/server: No such file or directory
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# cd /vagrant/provision/server
-bash: cd: /vagrant/provision/server: No such file or directory
[root@server.bahis.net ~]# mkdir -p /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.bahis.net ~]# cp -R /etc/rsyslog.d/netlog-server.conf /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch netlog.sh
[root@server.bahis.net server]# chmod +x netlog.sh
[root@server.bahis.net server]# nano netlog.sh
```

Рисунок 12: Создание каталога netlog и копирование конфигурации rsyslog

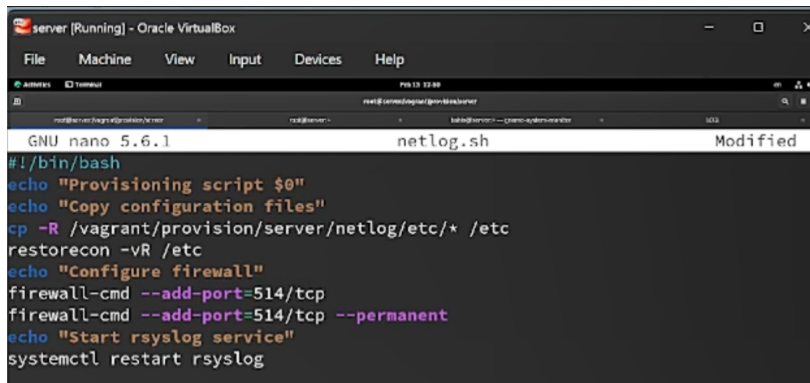
- Создан исполняемый скрипт `netlog.sh`



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing a file named "netlog.sh". The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/netlog/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=514/tcp
firewall-cmd --add-port=514/tcp --permanent
echo "Start rsyslog service"
systemctl restart rsyslog
```

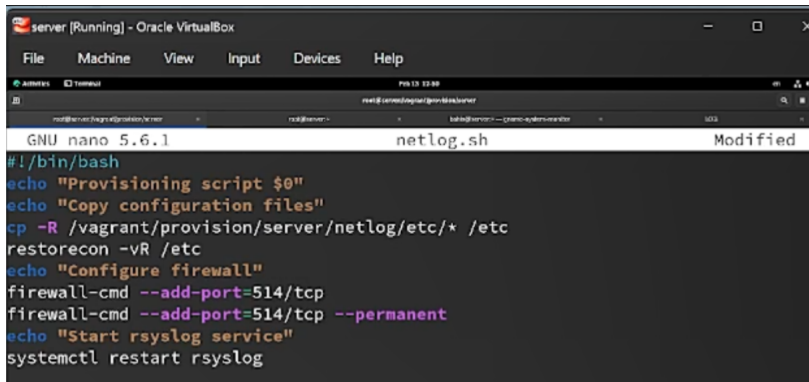
- Создан исполняемый скрипт `netlog.sh`
- Назначены права выполнения



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing the file `netlog.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/netlog/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=514/tcp
firewall-cmd --add-port=514/tcp --permanent
echo "Start rsyslog service"
systemctl restart rsyslog
```

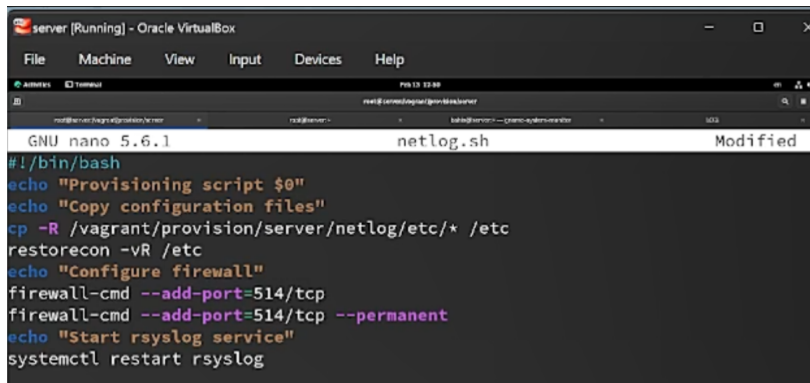
- Создан исполняемый скрипт `netlog.sh`
- Назначены права выполнения
- Добавлена автоматическая настройка



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing a file named `netlog.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/netlog/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=514/tcp
firewall-cmd --add-port=514/tcp --permanent
echo "Start rsyslog service"
systemctl restart rsyslog
```

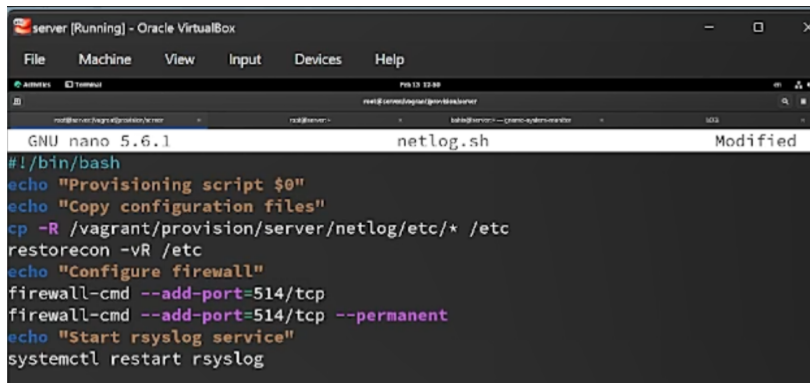
- Создан исполняемый скрипт `netlog.sh`
- Назначены права выполнения
- Добавлена автоматическая настройка
- Копирование файлов в `/etc`



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing the file `netlog.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/netlog/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=514/tcp
firewall-cmd --add-port=514/tcp --permanent
echo "Start rsyslog service"
systemctl restart rsyslog
```

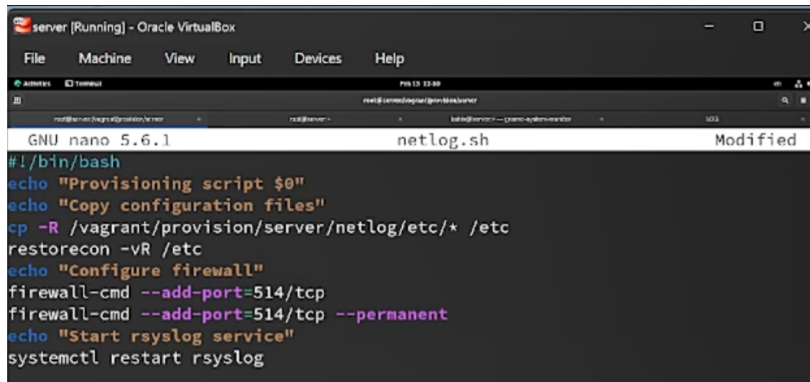
- Создан исполняемый скрипт `netlog.sh`
- Назначены права выполнения
- Добавлена автоматическая настройка
- Копирование файлов в `/etc`
- Восстановление контекстов SELinux



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing a file named `netlog.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/netlog/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=514/tcp
firewall-cmd --add-port=514/tcp --permanent
echo "Start rsyslog service"
systemctl restart rsyslog
```

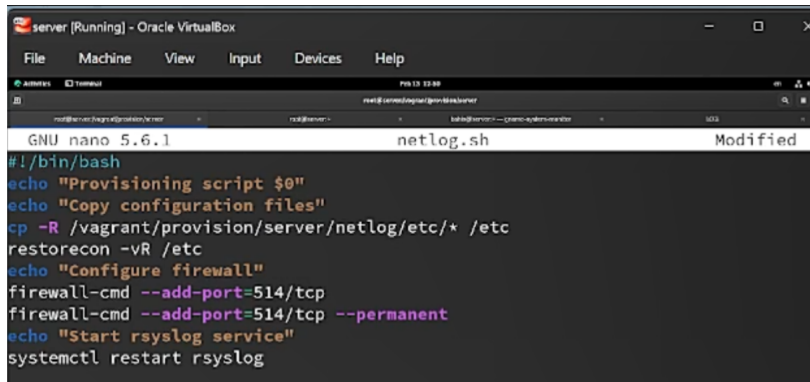
- Создан исполняемый скрипт `netlog.sh`
- Назначены права выполнения
- Добавлена автоматическая настройка
- Копирование файлов в `/etc`
- Восстановление контекстов SELinux
- Открытие порта 514



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing a file named `netlog.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/netlog/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=514/tcp
firewall-cmd --add-port=514/tcp --permanent
echo "Start rsyslog service"
systemctl restart rsyslog
```

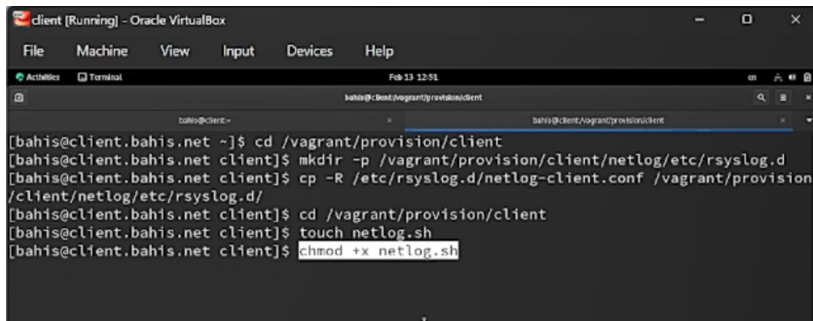
- Создан исполняемый скрипт `netlog.sh`
- Назначены права выполнения
- Добавлена автоматическая настройка
- Копирование файлов в `/etc`
- Восстановление контекстов SELinux
- Открытие порта 514
- Перезапуск `rsyslog`



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing a file named `netlog.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/netlog/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=514/tcp
firewall-cmd --add-port=514/tcp --permanent
echo "Start rsyslog service"
systemctl restart rsyslog
```

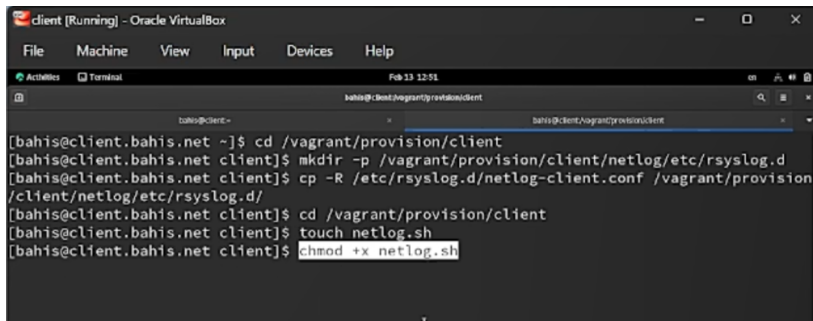
- Переход в `/vagrant/provision/client`



```
client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 12:51
bahis@client:/vagrant/provision/client
bahis@client~$ cd /vagrant/provision/client
bahis@client.bahis.net client]$ mkdir -p /vagrant/provision/client/netlog/etc/rsyslog.d
bahis@client.bahis.net client]$ cp -R /etc/rsyslog.d/netlog-client.conf /vagrant/provision/client/netlog/etc/rsyslog.d/
bahis@client.bahis.net client]$ cd /vagrant/provision/client
bahis@client.bahis.net client]$ touch netlog.sh
bahis@client.bahis.net client]$ chmod +x netlog.sh
```

Рисунок 14: Создание каталога netlog и копирование конфигурации клиента

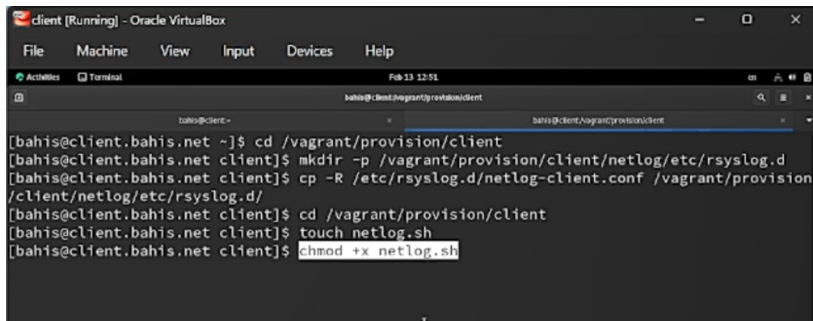
- Переход в `/vagrant/provision/client`
- Создание каталога `netlog/etc/rsyslog.d`



```
client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 12:51
bahis@client:/vagrant/provision/client
bahis@client:~$ cd /vagrant/provision/client
bahis@client.bahis.net client]$ mkdir -p /vagrant/provision/client/netlog/etc/rsyslog.d
bahis@client.bahis.net client]$ cp -R /etc/rsyslog.d/netlog-client.conf /vagrant/provision/client/netlog/etc/rsyslog.d/
bahis@client.bahis.net client]$ cd /vagrant/provision/client
bahis@client.bahis.net client]$ touch netlog.sh
bahis@client.bahis.net client]$ chmod +x netlog.sh
```

Рисунок 14: Создание каталога netlog и копирование конфигурации клиента

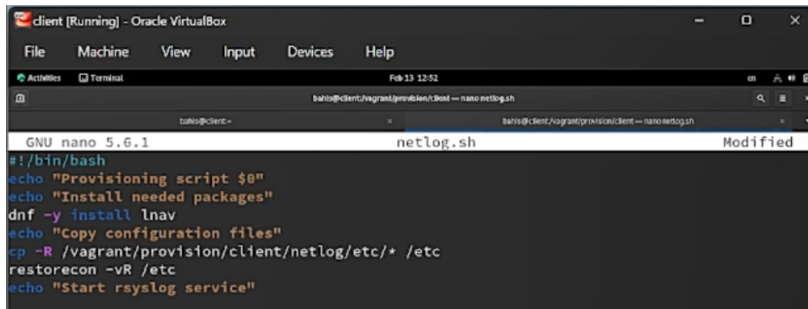
- Переход в `/vagrant/provision/client`
- Создание каталога `netlog/etc/rsyslog.d`
- Копирование `netlog-client.conf`



```
client [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Feb 13 12:51
bahis@client:/vagrant/provision/client
bahis@client:~$ cd /vagrant/provision/client
bahis@client.bahis.net client]$ mkdir -p /vagrant/provision/client/netlog/etc/rsyslog.d
bahis@client.bahis.net client]$ cp -R /etc/rsyslog.d/netlog-client.conf /vagrant/provision/client/netlog/etc/rsyslog.d/
bahis@client.bahis.net client]$ cd /vagrant/provision/client
bahis@client.bahis.net client]$ touch netlog.sh
bahis@client.bahis.net client]$ chmod +x netlog.sh
```

Рисунок 14: Создание каталога netlog и копирование конфигурации клиента

- Создан исполняемый файл `netlog.sh`

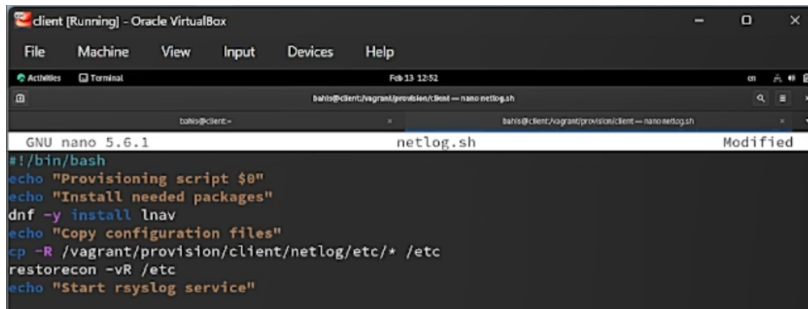


The screenshot shows a terminal window titled "client [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". Below the menu bar, there is a status bar showing "Feb 13 12:52" and "on". The terminal content shows the command `bahts@client:/vagrant/provision/client -- nano netlog.sh` being executed. The terminal output shows the content of the `netlog.sh` script, which is a bash script that provisions the system. The script content is as follows:

```
GNU nano 5.6.1 netlog.sh Modified
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install lnav
echo "Copy configuration files"
cp -R /vagrant/provision/client/netlog/etc/* /etc
restorecon -vR /etc
echo "Start rsyslog service"
```

Рисунок 15: Содержимое скрипта `netlog.sh` на клиенте

- Создан исполняемый файл `netlog.sh`
- Добавлен сценарий автоматической настройки клиента

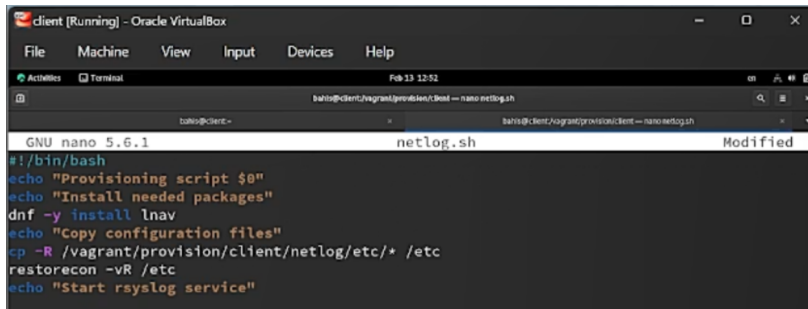


The screenshot shows a terminal window titled "client [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing the file `netlog.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $@"
echo "Install needed packages"
dnf -y install lnav
echo "Copy configuration files"
cp -R /vagrant/provision/client/netlog/etc/* /etc
restorecon -vR /etc
echo "Start rsyslog service"
```

Рисунок 15: Содержимое скрипта `netlog.sh` на клиенте

- Создан исполняемый файл `netlog.sh`
- Добавлен сценарий автоматической настройки клиента
- Установка `lnav`

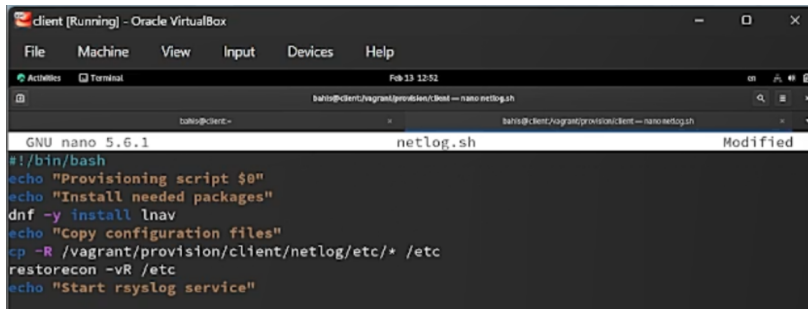


The screenshot shows a terminal window titled "client [Running] - Oracle VirtualBox". The terminal is running the `nano` text editor, editing the file `netlog.sh`. The script content is as follows:

```
GNU nano 5.6.1 netlog.sh Modified
#!/bin/bash
echo "Provisioning script $@"
echo "Install needed packages"
dnf -y install lnav
echo "Copy configuration files"
cp -R /vagrant/provision/client/netlog/etc/* /etc
restorecon -vR /etc
echo "Start rsyslog service"
```

Рисунок 15: Содержимое скрипта `netlog.sh` на клиенте

- Создан исполняемый файл `netlog.sh`
- Добавлен сценарий автоматической настройки клиента
- Установка `lnav`
- Копирование файлов в `/etc`

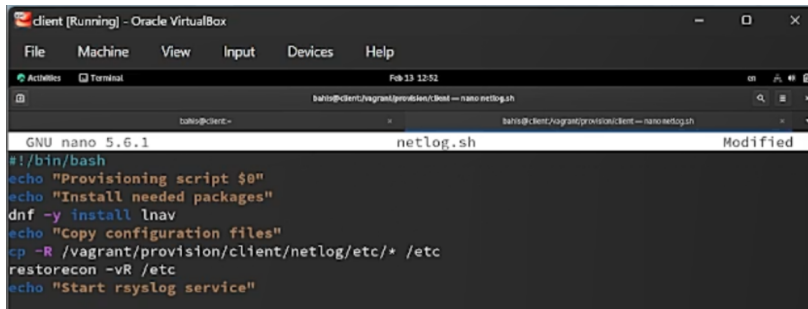


The screenshot shows a terminal window titled "client [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing the file `netlog.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $@"
echo "Install needed packages"
dnf -y install lnav
echo "Copy configuration files"
cp -R /vagrant/provision/client/netlog/etc/* /etc
restorecon -vR /etc
echo "Start rsyslog service"
```

Рисунок 15: Содержимое скрипта `netlog.sh` на клиенте

- Создан исполняемый файл `netlog.sh`
- Добавлен сценарий автоматической настройки клиента
- Установка `lnav`
- Копирование файлов в `/etc`
- Восстановление контекстов SELinux

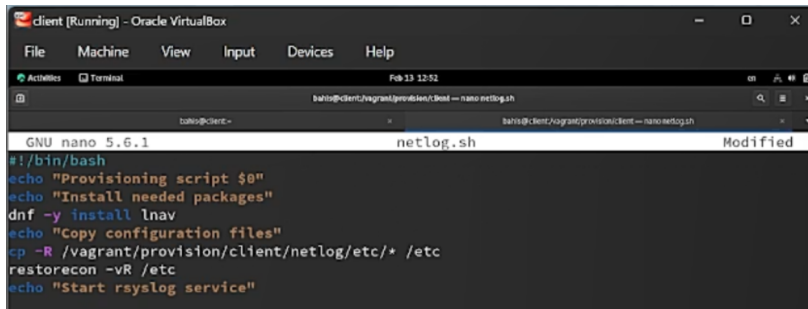


The screenshot shows a terminal window titled "client [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing the file `netlog.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $@"
echo "Install needed packages"
dnf -y install lnav
echo "Copy configuration files"
cp -R /vagrant/provision/client/netlog/etc/* /etc
restorecon -vR /etc
echo "Start rsyslog service"
```

Рисунок 15: Содержимое скрипта `netlog.sh` на клиенте

- Создан исполняемый файл `netlog.sh`
- Добавлен сценарий автоматической настройки клиента
- Установка `lnav`
- Копирование файлов в `/etc`
- Восстановление контекстов SELinux
- Перезапуск `rsyslog`



The screenshot shows a terminal window titled "client [Running] - Oracle VirtualBox". The terminal is running the `nano` text editor, editing the file `netlog.sh`. The script content is as follows:

```
GNU nano 5.6.1 netlog.sh Modified
#!/bin/bash
echo "Provisioning script $@"
echo "Install needed packages"
dnf -y install lnav
echo "Copy configuration files"
cp -R /vagrant/provision/client/netlog/etc/* /etc
restorecon -vR /etc
echo "Start rsyslog service"
```

Рисунок 15: Содержимое скрипта `netlog.sh` на клиенте

- В Vagrantfile добавлен блок provisioning для сервера

```
server.vm.provision "server netlog",  
type: "shell",  
preserve_order: true,  
path: "provision/server/netlog.sh"
```

Рисунок 16: Добавление provisioning для сервера в Vagrantfile

- В Vagrantfile добавлен блок provisioning для сервера
- Указан скрипт provision/server/netlog.sh

```
server.vm.provision "server netlog",  
type: "shell",  
preserve_order: true,  
path: "provision/server/netlog.sh"
```

Рисунок 16: Добавление provisioning для сервера в Vagrantfile

- В Vagrantfile добавлен блок provisioning для сервера
- Указан скрипт provision/server/netlog.sh
- Сохранена последовательность выполнения

```
server.vm.provision "server netlog",  
type: "shell",  
preserve_order: true,  
path: "provision/server/netlog.sh"
```

Рисунок 16: Добавление provisioning для сервера в Vagrantfile

- Добавлен аналогичный блок для клиента

```
client.vm.provision "client netlog",  
type: "shell",  
preserve_order: true, I  
path: "provision/client/netlog.sh"
```

Рисунок 17: Добавление provisioning для клиента в Vagrantfile

- Добавлен аналогичный блок для клиента
- Указан скрипт `provision/client/netlog.sh`

```
client.vm.provision "client netlog",  
type: "shell",  
preserve_order: true, I  
path: "provision/client/netlog.sh"
```

Рисунок 17: Добавление provisioning для клиента в Vagrantfile

- Добавлен аналогичный блок для клиента
- Указан скрипт `provision/client/netlog.sh`
- Обеспечена автоматическая настройка при запуске VM

```
client.vm.provision "client netlog",  
type: "shell",  
preserve_order: true, I  
path: "provision/client/netlog.sh"
```

Рисунок 17: Добавление provisioning для клиента в Vagrantfile

Раздел 3

3. Выводы

3.1 Выводы

- Настроено централизованное журналирование на базе `rsyslog`

3.1 Выводы

- Настроено централизованное журналирование на базе `rsyslog`
- Организован приём журналов по TCP 514 на сервере

3.1 Выводы

- Настроено централизованное журналирование на базе `rsyslog`
- Организован приём журналов по TCP 514 на сервере
- Реализовано перенаправление сообщений с клиента

3.1 Выводы

- Настроено централизованное журналирование на базе `rsyslog`
- Организован приём журналов по TCP 514 на сервере
- Реализовано перенаправление сообщений с клиента
- Подтверждена корректность работы через `/var/log/messages` и `lnav`

3.1 Выводы

- Настроено централизованное журналирование на базе rsyslog
- Организован приём журналов по TCP 514 на сервере
- Реализовано перенаправление сообщений с клиента
- Подтверждена корректность работы через `/var/log/messages` и `lnav`
- Реализована автоматизация через provisioning-скрипты

3.1 Выводы

- Настроено централизованное журналирование на базе rsyslog
- Организован приём журналов по TCP 514 на сервере
- Реализовано перенаправление сообщений с клиента
- Подтверждена корректность работы через `/var/log/messages` и `lnav`
- Реализована автоматизация через provisioning-скрипты
- Обеспечена воспроизводимость конфигурации

3.1 Выводы

- Настроено централизованное журналирование на базе rsyslog
- Организован приём журналов по TCP 514 на сервере
- Реализовано перенаправление сообщений с клиента
- Подтверждена корректность работы через /var/log/messages и lnav
- Реализована автоматизация через provisioning-скрипты
- Обеспечена воспроизводимость конфигурации
- Создана функционирующая система централизованного сбора журналов